

Question				Marking details			Marks available															
							AO1	AO2	AO3	Total	Maths	Prac										
1	(a)	(i)		<table><tr><td></td><td>α ketoglutaric</td><td>citric acid</td></tr><tr><td>Carbon</td><td>5</td><td>6</td></tr><tr><td>Hydrogen</td><td>6</td><td>8</td></tr><tr><td>Oxygen</td><td>5</td><td>7</td></tr></table> 1 mark for each column		α ketoglutaric	citric acid	Carbon	5	6	Hydrogen	6	8	Oxygen	5	7		2		2		
	α ketoglutaric	citric acid																				
Carbon	5	6																				
Hydrogen	6	8																				
Oxygen	5	7																				
		(ii)		Alpha ketoglutaric – B citric A Both required for 1 mark		1		1														
		(iii)		C + O → CO ₂ (1) / released as carbon dioxide (1) Ignore decarboxylation H + NAD→NADH ₂ / accepted by {NAD / cofactor / hydrogen acceptor} / {Reduces / combines} {NAD / cofactor / hydrogen acceptor} / Combines with NAD / carried to ETC (1) Ignore dehydrogenation		2		2														
	(b)	(i)		To show that reaction is enzymic / correct reference to dehydrogenase / owtte Ignore active yeast			1	1		1												

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		{In presence of malonic acid / it / tube 2} (methylene blue) {did not decolourise / stayed blue / is not reduced} (1) Showing it prevented {dehydrogenase activity / dehydrogenation} / owtte (1)			2	2		2
		(iii)		Any two (x1) from - Competitive Inhibitor (1) - Similar shape to succinic acid / {fits into / complementary to} active site(1) Reject Malonic acid is complementary to succinic acid - prevents the succinic acid from binding to the active site / forms an enzyme inhibitor complex / prevents the formation of an enzyme substrate complex (1)			2	2		2
				Question 1 total	0	5	5	10	0	5